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Leakage Insensitive Switch Control for Bandgap Thermal Sensors in Core-MOS Nodes

Abstract

Systems and methods are provided for a circuit comprising an operating voltage node, a current mirror circuit, and a dynamic element matching (DEM) circuit. The current mirror circuit is coupled to the operating voltage node and comprises a plurality of resistive devices. The current mirror circuit is configured to generate a bias current over the plurality of resistive devices. The DEM circuit comprises a plurality of DEM transistors coupled to the current mirror circuit. The DEM circuit is configured to switch the bias current through one or more of the plurality of DEM transistors. The DEM circuit includes a leakage reduction circuit configured to reduce a gate current of the one or more of the plurality of DEM transistors and is configured to generate a DEM output current based on the bias current and the gate current.

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Background/Summary

FIELD

[0001] The present disclosure relates to bandgap thermal sensors, and in particular to bandgap thermal sensors in core-MOS nodes.

BACKGROUND

[0002] A bandgap thermal sensor is a temperature sensor used in electronic equipment. In some bandgap thermal sensors, chopper switches are used to improve signal quality within the bandgap thermal sensors. Some chopper switches utilize dynamic element matching ("DEM") technology and are used in bandgap thermal sensors. Such devices can be susceptible to undesirable leakage currents.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] The following detailed description will be better understood when read in conjunction with the appended drawings. For the purpose of illustration, there is shown in the drawings certain embodiments of the present disclosure. It should be understood, however, that the invention is not limited to the precise arrangements and instrumentalities shown. The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate an implementation of systems and apparatuses consistent with the present invention and, together with the description, serve to explain advantages and principles consistent with the invention.

[0004] FIG. **1** depicts a block diagram of a bandgap thermal sensor, in accordance with some embodiments.

[0005] FIG. **2** depicts a detailed diagram of a current mirror circuit, a DEM circuit, and a cascode stage, in accordance with some embodiments.

[0006] FIG. **3** depicts a leakage reduction circuit **103** for one of the sets of resistive paths in accordance with an embodiment.

[0007] FIG. **4** a detailed diagram of a current mirror circuit, a DEM circuit, and a cascode stage, in accordance with some embodiments.

[0008] FIG. **5** depicts a block diagram of a bandgap thermal sensor with a leakage reduction chopper switch, in accordance with some embodiments.

[0009] FIG. **6** depicts a leakage reduction chopper switch, in accordance with some embodiments.

[0010] FIG. **7** depicts a leakage reduction switching element, in accordance with some embodiments.

[0011] FIG. **8** depicts a method of reducing gate leakage current, in accordance with some embodiments.

[0012] FIG. **9** depicts an alternative implementation of a leakage reduction circuit applied to the gate of a transistor on a resistive path of a DEM.

[0013] FIG. **10** depicts a second alternative implementation of a leakage reduction circuit applied to the drain/source of the transistor on a resistive path of a DEM.

[0014] FIG. **11** depicts a further example of the leakage reduction circuit of FIG. **10** in an implementation where no cascode stage is utilized.

[0015] FIG. **12** provides a further alternative for control of the gate of the first transistor between a current mirror transistor and a cascode transistor.

[0016] FIG. **13** provides an additional alternative for control of the gate of the first transistor between the current mirror transistor and the cascode transistor.

DETAILED DESCRIPTION

[0017] The following detailed description is provided to assist the reader in gaining a comprehensive understanding of the methods, apparatuses, and/or systems described herein. Accordingly, various changes, modifications, and equivalents of the systems, apparatuses and/or methods described herein will be suggested to those of ordinary skill in the art. Also, descriptions of well-known functions and constructions may be omitted for increased clarity and conciseness.
[0018] It is to be understood that the phraseology and terminology employed herein are for the purpose of description and should not be regarded as limiting. For example, the use of a singular term, such as, “a” is not intended as limiting of the number of items. Also the use of relational terms, such as but not limited to, “top,” “bottom,” “left,” “right,” “upper,” “lower,” “down,” “up,” “side,” are used in the description for clarity and are not intended to limit the scope of the invention or the appended claims. Further, it should be understood that any one of the features can be used separately or in combination with other features. Other systems, methods, features, and advantages of the invention will be or become apparent to one with skill in the art upon examination of the detailed description. It is intended that all such additional systems, methods, features, and advantages be included within this description, be within the scope of the present invention, and be protected by the accompanying claims.
[0019] As noted above, certain devices, such as BGT Bandgap/Thermal Sensor circuits may be susceptible to undesirable leakage currents, providing power inefficiencies. In some instances, core-MOS DEM chopper switches of BJT bandgap/thermal sensor (BG/TS) devices (e.g., 1.2V VDD devices) have evidenced undesirable gate leakages. Control of such leakages may be difficult in instances where a BJT BG/TS operates well with VDD>1V, which may be limited by P/N junction turn-on voltages and bias-current headroom in the VDD-minus-10% corner. Further, certain technologies such as Core-MOS may operate best with V.sub.gd/V.sub.gs/V.sub.ds less than 1V for reliability considerations. DEM/chopper switches may be used in BJT BG/TS to reduce mismatch. However, reliability concerns may remain in certain instances, such as core-MOS-only technologies.
[0020] Systems and methods herein, in embodiments, provide designs that may operate in core-MOS-only technologies, such that node scaling is maintained. In embodiments, the temperature sensing device (BJT) and main circuit architectures can be implemented away from BG/TS architectures, which can reduce node-to-node design migration efforts. Systems and methods, in embodiments, are proposed to limit the impact of gate leak, which may be applicable to certain technologies such as a pure core-MOS DEM/chopper circuit in HV design, as well as others.
[0021] FIG. **1** depicts a block diagram of a bandgap thermal sensor, in accordance with some embodiments. The bandgap thermal sensor **100** is responsive to an operating voltage node (not shown) and produces a signal that is indicative of a temperature. In the example shown in FIG. **1**, the bandgap thermal sensor **100** includes a current mirror circuit **101**, a dynamic element matching (DEM) circuit **102**, and a cascode stage **104** that provide an output to a chopper circuit **107** from which the temperature measurement can be ascertained.
[0022] The current mirror circuit **101** is coupled to the operating voltage node (not shown). The current mirror circuit **101** includes a plurality of current matching devices (e.g., transistors) **232**, **234**, **236**, **238**. The current mirror circuit **101** generates one or more bias currents **105**, **244**, **246**, **248**, one across each of the plurality of the current matching devices **232**, **234**, **236**, **238**.
[0023] The DEM circuit **102** is configured to selectively transfer each bias current **105** over one or more resistive devices (e.g., transistors) based on, for example, characteristics of the resistive devices or magnitude of the bias current **105**. The DEM circuit **102** includes a leakage reduction circuit **103**. The leakage reduction circuit **103** is coupled to the plurality of resistive devices and is configured to reduce an amount of leakage current flowing from the resistive devices to an external node (e.g., ground). The DEM circuit **102** generates a DEM output current **106**.
[0024] The cascode stage **104** receives the DEM output current **106** at a plurality of resistive devices connected in parallel. The cascode stage **104** generates a cascode stage output current **108**.

The cascode stage output current **108** is received by the chopper circuit **107**. The chopper circuit **107** is used to control an amount of current passing through a resistive element (e.g., a resistor). A temperature can be determined based on the current passing through the resistive element within the chopper circuit **107**.

[0025] FIG. **2** depicts a diagram of a current mirror circuit, a DEM circuit, and a cascode stage, in accordance with some embodiments. In the example shown in FIG. **2**, each of a plurality of current mirror current matching devices **232**, **234**, **236**, **238** provides a bias current, **105**, **244**, **246**, **248**, respectively. In an embodiment, the bias currents **105**, **244**, **246**, **248** are proportional to one another (e.g., the second bias current **244** is about K-times the magnitude of the first bias current **105**). The bias currents **105**, **244**, **245**, **246** are received by the DEM circuit **102**. The DEM circuit **102** is configured to switch each bias current **105**, **244**, **245**, **246** through one or more of a plurality of resistive paths. For example, for bias current **105**, the DEM circuit **102** includes a resistive path including a first transistor **201**, a resistive path including a second transistor **202**, a resistive path including a third transistor **203**, and a resistive path including a fourth transistor **204**. In operation, one or more of those transistors **201**, **202**, **203**, **204** may be selective as operative at a time. In embodiments, the resistive path selected by the DEM circuit **102** may be selected based on a digital control signal applied at the gates of those transistors **201**, **202**, **203**, **204**. In the example of FIG. **2**, the resistive path transistors **201**, **202**, **203**, **204** are active low transistors, where a ground signal applied at one of p**1**, p**2**, p**3**, p**4** will activate the respective transistor **201**, **202**, **203**, **204**. In embodiments each of the resistive paths (16 resistive paths in FIG. **2**) is independently controlled. In other examples, fewer signals are used. In the example of FIG. **2**, four control signals (p**1**, p**2**, p**3**, p**4**) are utilized and provide control to four resistive path transistors each.

[0026] The cascode stage **104** includes a plurality of cascode transistors **215**, **216**, **217**, **218** connected in parallel for each of the plurality of current branches. The input to those cascode transistors **215**, **216**, **217**, **218** is dependent on which of the resistive path transistors (e.g., transistors **201**, **202**, **203**, **204** in the first set of resistive paths) is active as commanded by its corresponding control signal. In the example of FIG. **2**, the first cascode transistor **215** receives current from the first resistive path of each of the four sets, the second cascode transistor **266** receives current from the second resistive path of the four sets, the third cascode transistor **217** receives current from the third resistive path of the four sets, and the fourth cascode transistor **218** receives current from the fourth resistive path of the four sets.

[0027] Each of the sets of resistive paths are responsive at the gates of their controlling transistors (e.g., at the gates of transistors **208**, **210**, **212**, **214** in the first set of resistive paths) to a leakage reduction circuit **103**, which facilitates control of signals p**1**, p**2**, p**3**, p**4**. FIG. **3** depicts a leakage reduction circuit **103** for one of the sets of resistive paths in accordance with an embodiment. In the example of FIG. **3**, the leakage reduction circuit **103** provides two transistors responsive to the control nodes p**1**, p**2**, p**3**, p**4**. Specifically, the leakage reduction circuit **103** includes a first operating transistor **207** coupled to the first transistor **201** at p**1** and a first ground transistor **208** coupled to the first transistor **201** at p**1**. Specifically, the first operating transistor **207** is a PMOS transistor that includes a gate terminal that is coupled to an operating voltage node **205** when the bias current **105** passes through the first transistor **201**. The operating voltage node **205** may have a voltage of, for example, 1.2 V. The first operating transistor **207** further includes a source terminal that is coupled to a gate terminal of the first transistor. The first ground transistor **208** includes a gate terminal that is coupled to a ground voltage node **206** when the bias current **105** passes through the first transistor **201**. The first ground transistor **208** further includes a drain terminal that is coupled to the gate terminal of the first transistor **201**.

[0028] The leakage reduction circuit **103** further includes a second operating transistor **209**, a third operating transistor **211**, and a fourth operating transistor **213** that are responsive to control nodes p**2**, p**3**, and p**4**, respectively. Each of the second operating transistor **209**, the third operating transistor **211**, and the fourth operating transistor **213** include a source terminal that is coupled to a

gate terminal of the second transistor **202**, the third transistor **203**, and the fourth transistor **204**, respectively. The leakage reduction circuit **103** further includes a second ground transistor **210**, a third ground transistor **212**, and a fourth ground transistor **214**, each responsive to control nodes p**2**, p**3**, and p**4**, respectively. Each of the second ground transistor **210**, the third ground transistor **212**, and the fourth ground transistor **214** include a drain terminal that is coupled to the gate terminal of the second transistor **202**, the third transistor **203**, and the fourth transistor **204**, respectively.
[0029] In the example of FIG. **3**, signal p**1** is active, as indicated by the ground signal **206** at transistor **208**, while signals p**2**, p**3**, p**4** are inactive, as indicated by the high signals at the gates of transistors **210**, **212**, **214**. This results in bias voltage **105** traversing the first transistor **201**. When the bias current **105** passes over the first transistor **201**, the leakage reduction circuit **103** thus forms a stacked-gate structure including the first transistor **201**, the first ground transistor **208**, and the cascode transistor **215** connected to the first transistor **201**. The leakage reduction circuit thus reduces the gate-to-source voltage of the first transistor **201** when the bias current passes over the first transistor **201**. Accordingly, the gate current **216** of the first transistor **201** is negligible, and substantially all of the bias current **105** delivered to the DEM circuit **102** is received by the cascode stage **104**. In a similar way, the leakage reduction circuit **103** can reduce the gate-to-source voltage of the second transistor **202**, the third transistor **203**, or the fourth transistor **204** based on which of transistors **201**, **202**, **203**, **304** over which the bias current **105** is flowing, as controlled by the signals applied to p**1**, p**2**, p**3**, p**4** via transistors **208**, **210**, **212**, **214**.
[0030] Where the first transistor **201** is selected via an active p**1** signal and the other transistors **202**, **203**, **204** are not, current **105** does not traverse those other transistors **202**, **203**, **204**. When the bias current **105** is not passing over the second transistor **202**, the gate terminal of the second operating transistor **209** is coupled to the ground voltage node **206** and the gate terminal of the second ground transistor **210** is coupled to the operating voltage node **205**. Similarly, the gate terminals of the third operating transistor **211** and the fourth operating transistor **213** are coupled to the ground terminal node **206** when the bias current **105** is not passing through the third transistor **203** or the fourth transistor **204**, and the gate terminals of the third ground transistor **212** and the fourth ground transistor **214** are coupled to the operating voltage node **205** when the bias current **105** is not passing through the third transistor **203** or the fourth transistor **204**.
[0031] FIG. **4** a detailed diagram of a current mirror circuit, a DEM circuit, and a cascode stage, in accordance with some embodiments. In the example of FIG. **4**, details of the leakage reduction circuit **103** associated with the first set of resistive paths associated with the first bias current **205** are shown in detail via transistors **207**, **208**, **209**, **210**, **211**, **212**, **213**, and **214**, their connection to respective ones of transistors **201**, **202**, **203**, **204**, and connections to the gates of the cascode transistors **215**, **216**, **217**, **218**. Details of the leakage reduction circuits **103** associated with the second, third, and fourth sets of resistive paths are presented in simplified states.
[0032] FIG. **5** depicts a block diagram of a bandgap thermal sensor with a leakage reduction chopper switch, in accordance with some embodiments. The bandgap thermal sensor **300** includes a current generator **302** including a current source **304** and a plurality of current generator transistors **305**. There may be a constant number K of current generator transistors **305**. The current source **304** generates the bias current **105**, which is received by the current mirror circuit **101**. The current mirror circuit **101** generates a DEM current **307** equal to the constant K times the bias current **105**. The current mirror circuit **101** produces the bias current **105** across the remainder of a plurality of current mirror transistors. As described above, the DEM circuit **102** selectively switches the bias current **105** through one or more of a plurality of DEM transistors. The cascode stage **104** includes a plurality of cascode transistors connected in parallel. The cascode transistors include a first cascode transistor **308** that receives the DEM current **307** and a second cascode transistor **309** that receives the bias current **105**.
[0033] The first cascode transistor **308** is coupled to a first end of a first chopper switch **311** and the second cascode transistor **309** is coupled to a second end of the first chopper switch **311**. An output

of the first chopper switch **311** is coupled to a second chopper switch **303**. The first cascode transistor is further coupled to a first temperature sensing transistor **312**. The second cascode transistor **309** is further coupled to a first end of a resistor **310**. A second end of the resistor **310** is coupled to a second temperature sensing transistor **313**. The first temperature sensing transistor **312** and the second temperature sensing transistor **313** are coupled to a ground voltage node **314**. The temperature of the bandgap thermal sensor **300** can be calculated based on comparing a voltage appearing across the first temperature sensing transistor **312** with a voltage appearing across the second temperature sensing transistor **313**. The operation of the first chopper switch **311** and the second chopper switch **303** are described further in the description of FIG. **6**.

[0034] FIG. **6** depicts a leakage reduction chopper switch, in accordance with some embodiments. The leakage reduction chopper switch **400** may be, for example, the first chopper switch **311** or the second chopper switch **303** shown in FIG. **5**. The leakage reduction chopper switch **400** includes a positive input voltage node **401**, a negative input voltage node **402**, a positive output voltage node **403**, and a negative output voltage node **404**. The leakage reduction chopper switch **400** further includes a first switching element **407**, a second switching element **408**, a third switching element **409**, and a fourth switching element **410**. The first switching element **407** is used to connect and disconnect the positive input voltage node **401** to the positive output voltage node **403**. The second switching element **408** is used to connect and disconnect the positive input voltage node **401** to the negative output voltage node **404**. The third switching element **409** is used to connect and disconnect the negative input voltage node **402** from the positive output voltage node **403**. The fourth switching element **410** is used to connect and disconnect the negative input voltage node **402** from the negative output voltage node **404**. Each of the switching elements **407**, **408**, **409**, **410** is controlled by a digital operating voltage DVDD **405** and a source voltage VS **406**. The operating of each of the switching elements **407**, **408**, **409**, **410** is described further in the description of FIG. **7**.

[0035] FIG. **7** illustrates a leakage reduction switching element, in accordance with some embodiments. The leakage reduction switching element **500** may be used as any of the switching elements **407**, **408**, **409**, **410** in the leakage reduction chopper switch **400** shown in FIG. **6**. In the example shown in FIG. **7**, the leakage reduction switching element **500** includes a buffer **501**. The buffer **501** receives a switching element input signal **506**. The switching element input signal **506** may either be the digital operating voltage DVDD **405** or the source voltage VS **406**. After passing through the buffer **501**, the switching element input signal **506** is received at a gate terminal of a PMOS transistor **502** and a gate terminal of an NMOS transistor **503**.

[0036] If the switching element input signal **506** is sufficiently high (e.g., the digital operating voltage DVDD **405**), the NMOS transistor **503** is enabled and the ground voltage **314** (e.g., 0 volts) is coupled to a switching element output node **504**, which can correspond, for example, to a switching element being closed. The leakage reduction switching element **500** includes a voltage source **505** that generates the source voltage VS **406** between the ground voltage node **314** and the NMOS transistor **503**. If the switching element input signal **506** is sufficiently low (e.g., the source voltage VS **406**), the PMOS transistor **502** is enabled and the digital operating voltage DVDD **405** is coupled to the switching element output node **504**, which can correspond, for example, to a switching element being opened. Those signals are then applied as shown to one of the switching elements (e.g., switching element **410** of FIG. **6**) so as to control that switch with minimized gate leak.

[0037] FIG. **8** depicts a method of reducing gate leakage current, in accordance with some embodiments. In the example shown in FIG. **8**, the method **600** includes a first step **601** of receiving a digital control signal. The method **600** further includes a second step **602** of distributing the bias current through one or more of a plurality of transistors based on the digital control signal. The method **600** further includes a third step **603** of coupling the one or more transistors to an operating transistor or a ground transistor based on the digital control signal. The coupling reduces a gate-to-source voltage of the one or more transistors.

[0038] Leakage reduction circuits, as described herein may take a variety of forms. FIG. **9** depicts an alternative implementation of a leakage reduction circuit applied to the gate of a transistor on a resistive path of a DEM. A bias current **105** is received from the first current matching device **232** of the current mirror **101** at a first transistor **201**. The output of that transistor is provided to a first transistor **215** of the cascode stage **104**. The gate of the first transistor **201** is controlled by a leakage reduction circuit transistor **702**. When the first transistor **201** is selected as active, the leakage reduction circuit transistor **702** pulls node p**1** low (as shown in FIG. **9**), which allows the bias current to flow from the current mirror transistor **232** to the cascode transistor **215**. In this on state, the inversion gate leak (I.sub.GI) from the bias current to the grounded nodes of the leakage reduction circuit transistor **702** is limited by a small V.sub.GS associated with the first transistor **201**. When the first transistor **201** is not selected, a high signal is applied to node p**1**, turning the first transistor **201** off, preventing the flow of current from the current mirror transistor **232** to the cascode transistor **215**. In this instance, gate leak from the high signal at p**1** through the first switch **201** is limited (e.g., is substantially less than I.sub.GI).
[0039] FIG. **10** depicts a second alternative implementation of a leakage reduction circuit applied to the drain/source of the transistor **201** on a resistive path of a DEM. In this instance, the bias current **105** from the first current matching device **232** is directly received by the first transistor **215** of the cascode stage **104**. The gate of the first transistor **215** of the cascode stage **104** is controlled by the first transistor **201** and the leakage reduction circuit transistor **802**. The leakage reduction current transistor **802** receives the inverse (p**1**_b) of the control signal applied at node p**1**, controlling a signal provided to the gate of the cascode transistor **215**. When a low signal is applied at p**1**, the first transistor **201** is on, providing a path from **804** to the gate of the cascode transistor **215**. This pulls the gate of the cascode transistor **204** low, allowing the bias voltage **105** to traverse that transistor **215**. Correspondingly, a high signal is applied at p**1**_b, turning off leakage reduction circuit transistor **802**. When a high signal is applied to the first transistor **201** at p**1**, a low signal is provided to the gate of the leakage reduction circuit transistor **802**, applying a high signal to the gate of the cascode transistor **215**, turning it off.
[0040] In this example, when the first transistor **201** is on, V.sub.gs,sw ˜ (V.sub.DD−V.sub.dsat(current_mirror_transistor))−V.sub.gs,cascode, where the switch gate leak is not on the bias current path. When the first transistor **201** is off, the cascode-gate is tied to VDD. As depicted in the following figure, a similar implementation can be applied directly to a current mirror transistor.
[0041] FIG. **11** depicts a further example of the leakage reduction circuit of FIG. **10** in an implementation where no cascode stage is utilized. Here, the gate of the first current matching device **232** of the current mirror circuit **101** is controlled directly. A low signal at p**1** turns the first transistor **201**, such that a path from **804** to the to the gate of the current mirror transistor **232** is present. This pulls the gate of the current mirror transistor **232** low, allowing the bias voltage **105** to traverse that transistor **232**. When a high signal is applied to the first transistor **201** at p**1**, a high signal from the leakage reduction circuit transistor **802** is applied, turning the current mirror transistor **223** off.
[0042] FIG. **12** provides a further alternative for control of the gate of the first transistor **201** between a current mirror transistor **232** and a cascode transistor **215**. There, when the first transistor **201** is on, V.sub.gs,sw ˜ (V.sub.DD−V.sub.dsat(current_mirror_transistor))−V.sub.gs,sw1. When the switch is off, the gate of the cascode transistor **215** is tied to VDD. FIG. **13** provides an additional alternative for control of the gate of the first transistor between the current mirror transistor **232** and the cascode transistor **215**. There, when the switch is on, V.sub.gs,sw ˜ (V.sub.DD−V.sub.dsat(current_mirror_transistor))−V.sub.gs,sw1. And again, when the switch is off, the gate of the cascode transistor **215** is tied to V.sub.DD.
[0043] Systems and methods are described herein. In one example, a circuit comprises an operating voltage node, a current mirror circuit, and a dynamic element matching (DEM) circuit. The current

mirror circuit is coupled to the operating voltage node and comprises a plurality of resistive devices. The current mirror circuit is configured to generate a bias current over the plurality of resistive devices. The DEM circuit comprises a plurality of DEM transistors coupled to the current mirror circuit. The DEM circuit is configured to switch the bias current through one or more of the plurality of DEM transistors. The DEM circuit includes a leakage reduction circuit configured to reduce a gate current of the one or more of the plurality of DEM transistors and is configured to generate a DEM output current based on the bias current and the gate current.
[0044] In another example, a dynamic element matching (DEM) circuit comprises a plurality of DEM transistors. The DEM circuit is configured to receive a bias current and to selectively distribute the bias current using one or more of the plurality of DEM transistors based on a digital control signal. The DEM circuit further includes a leakage reduction circuit coupled to the plurality of DEM transistors. The leakage reduction circuit is configured to reduce a gate-to-source voltage of the one or more of the plurality of DEM transistors.
[0045] In another example, a method of reducing gate leakage current comprises receiving a digital control signal. The method further includes distributing a bias current through one or more of a plurality of transistors based on the digital control signal. The method further includes coupling the one or more transistors to an operating transistor or a ground transistor based on the digital control signal. The coupling reduces a gate-to-source voltage of the one or more transistors.
[0046] It will be appreciated by those skilled in the art that changes could be made to the embodiments described above without departing from the broad inventive concept thereof. It is understood, therefore, that the invention disclosed herein is not limited to the particular embodiments disclosed, and is intended to cover modifications within the spirit and scope of the present invention.

Claims

1. A circuit comprising: an operating voltage node; a current mirror circuit coupled to the operating voltage node, the current mirror circuit comprising a plurality of resistive devices, the current mirror circuit configured to generate a bias current over the plurality of resistive devices; and a dynamic element matching (DEM) circuit comprising a plurality of DEM transistors coupled to the current mirror circuit, the DEM circuit configured to switch the bias current through one or more of the plurality of DEM transistors, the DEM circuit including a leakage reduction circuit configured to reduce a gate current of the one or more of the plurality of DEM transistors, the DEM circuit configured to generate a DEM output current based on the bias current and the gate current.
2. The circuit of claim 1, further comprising a cascode stage coupled to the DEM circuit, the cascode stage including a plurality of cascode transistors configured to receive the DEM output current and to generate a cascode stage output current.
3. The circuit of claim 2, further comprising a chopper circuit coupled to the cascode stage, the chopper circuit configured to receive the cascode stage output current and to operate a chopper switch based on the cascode stage output signal.
4. The circuit of claim 1, wherein the DEM circuit includes a plurality of current branches, each of the current branches comprising the plurality of DEM transistors.
5. The circuit of claim 1, wherein the leakage reduction circuit includes an operating transistor and a ground transistor coupled to each of the DEM transistors.
6. The circuit of claim 1, wherein the leakage reduction circuit reduces the gate current of the one or more DEM transistors by reducing a gate-to-source voltage of the one or more DEM transistors.
7. The circuit of claim 1, wherein the leakage reduction circuit is configured to form a stacked gate structure including the one or more DEM transistors.
8. The circuit of claim 1, wherein the resistive devices are current mirror transistors.
9. A dynamic element matching (DEM) circuit comprising: a plurality of DEM transistors, the

DEM circuit configured to receive a bias current and to selectively distribute the bias current using one or more of the plurality of DEM transistors based on a digital control signal; and a leakage reduction circuit coupled to the plurality of DEM transistors, the leakage reduction circuit configured to reduce a gate-to-source voltage of the one or more of the plurality of DEM transistors.

10. The DEM circuit of claim 9, wherein the leakage reduction circuit is further configured to form a stacked gate structure including the one or more of the plurality of DEM transistors.

11. The DEM circuit of claim 9, wherein the leakage reduction circuit includes a separate operating transistor coupled to each of the plurality of DEM transistors.

12. The DEM circuit of claim 11, wherein the leakage reduction circuit includes a separate ground transistor coupled to each of the plurality of DEM transistors.

13. The DEM circuit of claim 12, wherein each of the operating transistors and each of the ground transistors are PMOS transistors.

14. The DEM circuit of claim 9, wherein the DEM circuit includes a plurality of current branches, each of the plurality of current branches including the plurality of DEM transistors.

15. The DEM circuit of claim 14, wherein each of the plurality of current branches is coupled to a current mirror transistor.

16. A method of reducing gate leakage current comprising: receiving a digital control signal; based on the digital control signal, distributing a bias current through one or more of a plurality of transistors; and based on the digital control signal, coupling the one or more transistors to an operating transistor or a ground transistor, the coupling reducing a gate-to-source voltage of the one or more transistors.

17. The method of claim 16, further comprising receiving the bias current and generating a DEM output current based on the bias current and the gate leakage current.

18. The method of claim 16, wherein the operating transistor is a PMOS operating transistor having a gate terminal coupled to an operating voltage node.

19. The method of claim 16, wherein the ground transistor is a PMOS ground transistor having a gate terminal coupled to a ground voltage node.

20. The method of claim 16, wherein the plurality of transistors are included in one of a plurality of current branches.